

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			straight line from anywhere along the top of Li bar to anywhere along the top of Cs bar – <i>ruler must be used</i> (1) potassium value further from the line than the sodium value (1) <i>accept numerical values for each metal</i>			2	2	2	
	(b)	(i)		similarity: observation (1) reason (1) both float densities less than water / less than 1 both form bubbles / fizz (hydrogen) gas formed both move (about)..... (hydrogen) gas formed do not accept: both melt / ball shape difference: observation (1) reason (1) potassium burns, lithium doesn't..... more exothermic / more reactive potassium moves about more more reactive potassium melts, lithium doesn't lithium has higher melting point neutral answer: potassium reacts more violently	4			4		4
		(ii)		reactants: $\text{Li} + \text{O}_2$ (1) product: Li_2O (1) balancing: $4\text{Li} + \text{O}_2 \rightarrow 2\text{Li}_2\text{O}$ (1) <i>only award balancing mark if the formulae of reactants and products are correct</i>		3		3	1	

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		(iii)		any three of following for (1) each - potassium's outer shell electron furthest away from nucleus - least attraction / electron most easily lost - chlorine's outer shell closest to nucleus - greatest attraction / electron most easily gained allow (1) for 'reactivity increases down Group 1 and decreases down Group 7' if no other mark awarded	3			3		
				Question 6 total	7	3	2	12	3	4